Data Classification Using AI

Project Overview

This project demonstrates a basic AI classification model using the Iris Dataset and the K-Nearest Neighbors (KNN) algorithm. The model classifies iris flowers into three species: Setosa, Versicolor, and Virginica based on sepal and petal measurements.

Dataset

- **Dataset:** Iris Dataset (150 samples)

- **Samples:** 120 training, 30 testing (80/20 split)

- **Features:** Sepal length, Sepal width, Petal length, Petal width

- **Classes:** Setosa, Versicolor, Virginica

Technologies Used

- Python, Scikit-Learn, NumPy, Pandas

- Algorithm: K-Nearest Neighbors (K=5)

- Data Scaling: StandardScaler

Evaluation Metrics

- Accuracy, Precision, Recall, F1 Score

- Confusion Matrix, Classification Report

Terminal Output

Below is the output generated after running **main.py**:

```
Dataset loaded successfully.
Flower classes: ['setosa' 'versicolor' 'virginica']
----------------------------------------
Training samples: 120
Testing samples: 30
----------------------------------------
Model Results
----------------------------------------
Accuracy: 1.0
Precision: 1.0
Recall   : 1.0
F1 Score : 1.0

Confusion Matrix
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Classification Report
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

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